

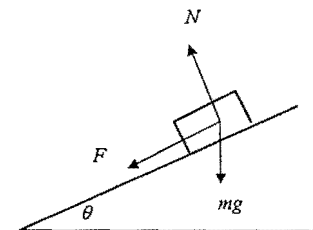
Question One (Start a new page)

Marks

- a) Consider the complex numbers $z_1 = \sqrt{2}(1 + i\sqrt{3})$ and $z_2 = 2\sqrt{6}(1 + i)$.
- i. Express $z = \frac{z_1}{z_2}$ in the form of $x + iy$, where x and y are real. 2
- ii. Write z_1 , z_2 and z in modulus/argument form. 5
- iii. Hence find the exact value of $\cos \frac{\pi}{12}$. 1
- b) Sketch on separate Argand diagrams the regions where
- i. $\operatorname{Re}(z + iz) \geq 2$ 2
- ii. $1 \leq |z - 1 - i| \leq 3$ where $z = x + iy$. 2
- c) By applying De Moivre's Theorem and by expanding $(\cos \theta + i \sin \theta)^5$, express $\sin 5\theta$ as a polynomial in $\sin \theta$. 3

Question Two (Start a new page)

- a) Given real positive numbers a , b and c such that $a > b > c$.
- i. Prove that $(a + b) > 2\sqrt{ab}$. 1
- ii. Show that $b^2 - a^2 < 2(b - a)\sqrt{ab}$. 1
- iii. Deduce that $(b - a)\sqrt{a} + (c - b)\sqrt{c} > \frac{c^2 - a^2}{2\sqrt{b}}$. 2
- b) i. Sketch the graph of $f(x) = 1 - \frac{9}{x^2} + \frac{18}{x^4}$, showing all stationary points and other essential features. 5
- ii. Hence find the set of values of the real numbers k such that the equations $f(x) = k$ has four distinct real roots. 1
- c) An object of mass m kg is travelling around a circular banked track of radius r metres and angle of banking θ . The mass is travelling at $v \text{ ms}^{-1}$. The forces acting on the object are the gravitational force mg newtons, a sideways friction force F newtons (acting down the road as shown) and a normal reaction N newtons to the road.

Diagram
not to scale

- i. By resolving forces vertically and horizontally, derive expressions for N and F . 4
- ii. Given the radius of the curve is 1 km and $\tan \theta = \frac{1}{100}$, find the velocity which will ensure no sideways friction. (Take $g = 10 \text{ ms}^{-2}$) 1

Question Three (Start a new page)

- a) Evaluate $\int_0^4 \frac{dx}{3 + \sqrt{x}}$. 3
- b) Let α , β and δ be the roots of $x^3 - x^2 + 2x - 1 = 0$.
- i. Find the value of $\alpha + \beta + \delta$. 1
- ii. Hence, or otherwise, find the cubic equation with roots: $-(\alpha + \beta)$, $-(\beta + \delta)$ and $-(\alpha + \delta)$. 2
- c) Consider the ellipse E with equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and its auxiliary circle C with equation $\frac{x^2}{16} + \frac{y^2}{16} = 1$.
- A straight line l parallel to the y axis, intersects the x axis at N and the curves E and C at the points P and Q respectively.
- Given that P and Q are both in the first quadrant and the coordinates of P on E are $(4 \cos \theta, 3 \sin \theta)$.
- i. Sketch the curves E and C showing the above information. 1
- ii. Write down the coordinates of N and Q in terms of θ . 2
- iii. Derive the equation of the tangent to the curve E at the point P . 2
- iv. Write down the equation of the tangent to the curve C at the point Q . 1
- v. The tangents at P and Q intersect at a point R . Show that R lies on the x axis. 2
- vi. Prove that $ON \cdot OR$ is independent of the positions P and Q . 1

Question Four (Start a new page)

Marks

a) Using the Table of Standard Integrals, find $\int \frac{1}{\sqrt{x^2 - 4x + 5}} dx$.

2

b) i. Find the constants A and B such that

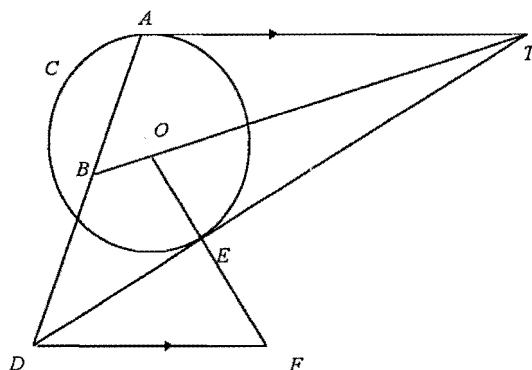
2

$$\frac{1}{\cos x} = \frac{A \cos x}{1 - \sin x} + \frac{B \cos x}{1 + \sin x}.$$

ii. Hence find the exact value of $\int_0^{\frac{\pi}{6}} \sec x \, dx$.

2

c) Diagram not to scale



In the above diagram, C is a circle with exterior point T . Tangents from T are drawn to meet C at the points A and E . The point O is the centre of C . The line BT passes through O . The line AD passes through B . The line OF passes through E . AT is parallel to DF .

i. Trace or copy the diagram onto your answer book and prove $\triangle OET \cong \triangle OAT$.

2

ii. Considering $\triangle OET$ and $\triangle DEF$, show that $DE = \frac{DF(ET^2 - OE^2)}{OT^2}$ by using double angle formula.

4

iii. Use the sine rule to show that $\frac{AB}{BD} = \frac{AT}{DT}$.

3

Question Five (Start a new page)

Marks

a) i. Prove $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

2

ii. Hence evaluate $\int_0^1 x(1-x)^n dx$.

3

b) Let $w = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$.

i Find all the complex roots of the equation $z^{10} - 1 = 0$ by writing them in terms of w and k , while k is a positive integer.

2

ii. Prove that $1 + w + w^2 + w^3 + \dots + w^9 = 0$.

1

iii. The quadratic equation $x^2 + bx + c = 0$, where b and c are real, has the root $w + w^4$. Find the other root in terms of w .

2

iv. Find b and express c in terms of $\sin \frac{\pi}{5}$.

5

Question Six (Start a new page)

a) Find $\int \frac{1}{e^x + e^{-x}} dx$.

3

b) Given that $I_n = \int_1^e (\ln y)^n dy$, $n = 0, 1, 2, 3, \dots$

i. Prove that $I_n = e - nI_{n-1}$

3

ii. Hence evaluate $\int_1^e (\ln y)^2 dy$.

2

c) The depth of water at the entrance to a harbour can be modeled using the equation $x = b + a \cos nt$ where x metres is the depth of water and t is time measured in hours. For a certain harbour, the first low tide for the day is at 5am and the water depth is 20m. The next high tide is $6\frac{1}{2}$ hours later and the corresponding depth is 28m.

i. Taking the first low tide for the day as the origin for measuring the time, write down the values of a , b and n .

2

ii. Find the depth of water at 9am. (correct to 3 significant figures)

2

iii. Find all the times after mid-night and before mid-day when water depth is 23 m. (correct to nearest minute)

2

iv. Find the greatest rate at which the tide is rising.

1

Question Seven (Start a new page)

Marks

- a) The circle $x^2 + y^2 = 9$ is rotated about the line $x = 8$ to form a torus. Using the method of cylindrical shells, find the volume of the torus. 4
- b) i. $xy = c^2$ is the result of rotating $x^2 - y^2 = a^2$ anticlockwise through an angle of 45° . Write down the relationship between a^2 and c^2 . 1
- ii. $P(x_1, y_1)$ is the point of intersection of the hyperbolas $xy = c^2$ and $x^2 - y^2 = a^2$ in the first quadrant. Prove that the tangent to $xy = c^2$ at the point P is $xy_1 + yx_1 = 2c^2$. 2
- iii. Write down the equation of the tangent to $x^2 - y^2 = a^2$ at the point P . 1
- iv. The tangent to the hyperbola $x^2 - y^2 = a^2$ at P meets its asymptotes at A and C while the tangent to the hyperbola $xy = c^2$ at P meets its asymptotes in B and D . 2
- v. Show that the co-ordinates of A are $(x_1 + y_1, x_1 + y_1)$. 3
- vi. Find the co-ordinates of B, C and D . 2
- vii. Prove that $ABCD$ is a square. 2

Question Eight (Start a new page)

- a) As shown in the diagram below, a light string of $2l$ metres long is attached to two points A and B . A mass of $3m$ kg is attached to the middle of the string and a second mass of m kg in the form of a ring is attached to the end of the string at B .

The $3m$ kg mass is rotating in circular motion at ω radians per second and the m kg mass is free to move up or down the smooth vertical rod AB . The string makes an angle θ with the vertical. (Assume the acceleration due to gravity is $g \text{ ms}^{-2}$).

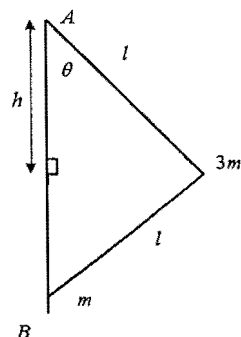


Diagram not to scale

- i. Given that h is the distance between A and the centre of the circular motion, find an expression for h in terms of g and ω . 4
- ii. If the $3m$ kg mass is replaced by a mass of m kg mass and the m kg ring is replaced by a ring of $3m$ kg, the speed of the rotating mass is doubled to 2ω radians per second. Determine if h is increased or decreased and give reasons. (note that $\omega > 1$) 3
- b) Mr Dud's crystal ball rests on a solid stand which is in the shape of a square based frustum as shown. 4

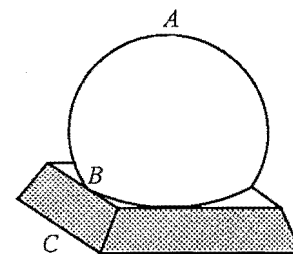
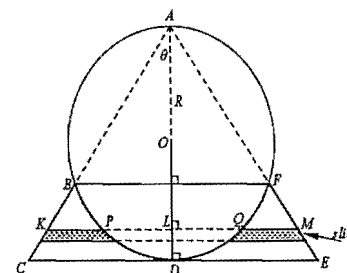


Diagram not to scale



cross-section

The stand is constructed so that the crystal ball of radius R fits snugly inside and just touches the centre of the square base. The side BC of the base slopes so that if extended it would pass through the top-most point of the ball at A and makes an angle θ with the vertical AD . Take O as the centre of the circle and let the distance OL be x units.

- i. Explain why $LQ = \sqrt{R^2 - x^2}$ and $LM = (R + x) \tan \theta$. 2
- ii. Consider a slice KLM of thickness Δx as shown perpendicular to AD . Show that it has a volume $\Delta V \approx \{4 \tan^2 \theta (R + x)^2 - \pi (R^2 - x^2)\} \Delta x$. 2
- iii. Find the volume of such a solid when the angle $\theta = \frac{\pi}{6}$. 4

END

MATHEMATICS Extension 2: Question...

Suggested Solutions

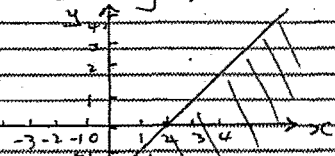
Marks

Marker's Comments

b) i) $\operatorname{Re}(z + iz) \geq 2$

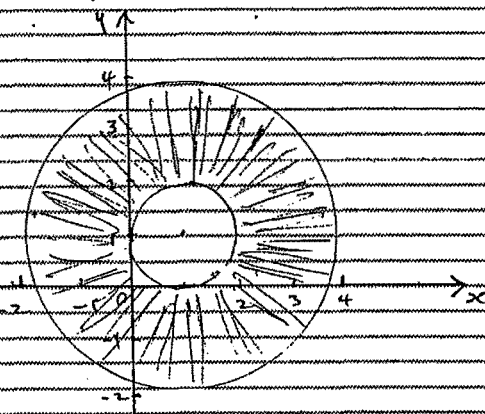
$\operatorname{Re}(x + iy + i(x + iy)) \geq 2$

$\therefore x - y \geq 2$



(ii) $1 \leq |z - (1 - i)| \leq 3$

$1 \leq |z - (1 + i)| \leq 3$



$(\cos 5\theta + i \sin 5\theta) = \cos 5\theta + i \sin 5\theta$ (De Moivre's theorem)

$= \cos^5 \theta + i 5 \cos^4 \theta \sin \theta + i^2 10 \cos^3 \theta \sin^2 \theta + i^3 10 \cos^2 \theta \sin^3 \theta + i^4 5 \cos \theta \sin^4 \theta + i^5 \sin^5 \theta$

Equate imaginary terms

$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$

$= 5(1 - \sin^2 \theta)^2 \sin \theta - 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta$

$= 5(1 - 2\sin^2 \theta + \sin^4 \theta) \sin \theta - 10\sin \theta + 10\sin^3 \theta + \sin^5 \theta$

$= 5\sin \theta - 10\sin^3 \theta + 5\sin^5 \theta - 10\sin \theta + 10\sin^3 \theta + \sin^5 \theta$

$= 16\sin^5 \theta - 20\sin^3 \theta + 5\sin \theta$

(2)

Graphs must be to scale

① inequality

② line $x - y = 2$

③ shading

(2)

① inside circle and shading

② outside circle

marks deducted for graphs not to scale or poorly drawn

(3)

①

①

①

①

①

①

①

MATHEMATICS Extension 2: Question...

Suggested Solutions

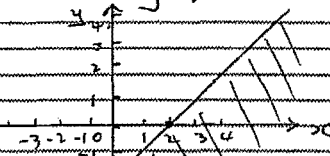
Marks

Marker's Comments

b) i) $\operatorname{Re}(z + iz) \geq 2$

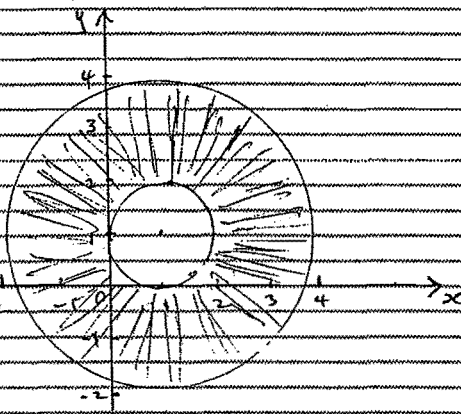
$\operatorname{Re}(x + iy + i(x + iy)) \geq 2$

$\therefore x - y \geq 2$



(ii) $1 \leq |z - (1 - i)| \leq 3$

$1 \leq |z - (1 + i)| \leq 3$



$(\cos 5\theta + i \sin 5\theta) = \cos 5\theta + i \sin 5\theta$ (De Moivre's theorem)

$= \cos^5 \theta + i 5 \cos^4 \theta \sin \theta + i^2 10 \cos^3 \theta \sin^2 \theta + i^3 10 \cos^2 \theta \sin^3 \theta + i^4 5 \cos \theta \sin^4 \theta + i^5 \sin^5 \theta$

Equate imaginary terms

$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$

$= 5(1 - \sin^2 \theta)^2 \sin \theta - 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta$

$= 5(1 - 2\sin^2 \theta + \sin^4 \theta) \sin \theta - 10\sin \theta + 10\sin^3 \theta + \sin^5 \theta$

$= 5\sin \theta - 10\sin^3 \theta + 5\sin^5 \theta - 10\sin \theta + 10\sin^3 \theta + \sin^5 \theta$

$= 16\sin^5 \theta - 20\sin^3 \theta + 5\sin \theta$

(2)

Graphs must be to scale

① inequality

② line $x - y = 2$

③ shading

(2)

① inside circle and shading

② outside circle

marks deducted for graphs not to scale or poorly drawn

(3)

①

①

①

①

①

①

①

Q2.

a) $a > b > c > 0 \Rightarrow \sqrt{a} > \sqrt{b} > \sqrt{c} > 0$

i) $(\sqrt{a} - \sqrt{b})^2 > 0$ (equality iff $a=b$)
 $a+b-2\sqrt{ab} > 0$
 $a+b > 2\sqrt{ab}$ #

ii) $b < a \therefore b-a < 0$

$(a+b)(b-a) < 2\sqrt{ab}(b-a)$

$b^2 - a^2 < 2\sqrt{ab}(b-a)$ # ①

iii) from ii since $c < b$

$\therefore c^2 - b^2 < 2\sqrt{bc}(c-b)$ ②

① + ② $c^2 - a^2 < 2\sqrt{ab}(b-a) + 2\sqrt{bc}(c-b)$

$c^2 - a^2 < 2\sqrt{b}[\sqrt{a}(b-a) + \sqrt{c}(c-b)]$

$\therefore \sqrt{a}(b-a) + \sqrt{c}(c-b) > \frac{c^2 - a^2}{2\sqrt{b}}$ #

b i) $f(x) = 1 - \frac{9}{x^2} + \frac{18}{x^4} = \frac{x^4 - 9x^2 + 18}{x^4} = \frac{(x^2-6)(x^2-3)}{x^4}$

$f'(x) = \frac{9x^2}{x^3} - \frac{72}{x^5} = 0$
 $f(x)=0$ when
 $x = \pm\sqrt{3}$
 $\pm\sqrt{6}$

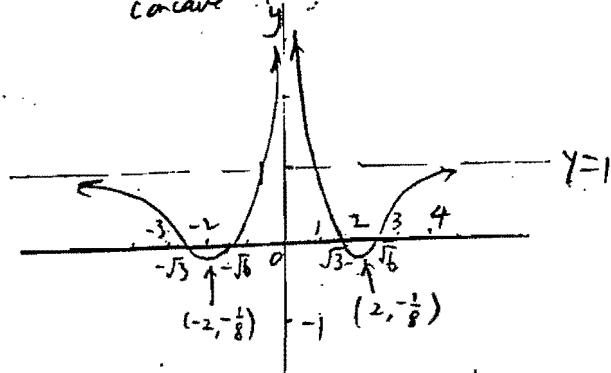
$\frac{18}{x^3} = \frac{72}{x^5} \Rightarrow x^2 = 4$ ($x \neq 0$)
 $x = \pm 2$ (S.P)

SP $x=2, y=-\frac{1}{8}$
 $x=-2, y=-\frac{1}{8}$

$f''(x) = \frac{-18 \times 3}{x^4} + \frac{72 \times 5}{x^6} = \frac{360}{x^6} - \frac{54}{x^4}$

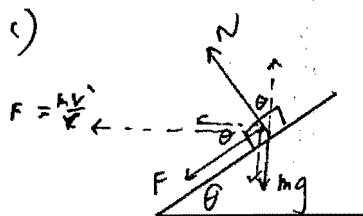
$f''(\pm 2) = \frac{360}{64} - \frac{54}{16} = \frac{90-54}{16} = \frac{36}{16} > 0$ (✓)

concave up min



when $-\frac{1}{8} < k < 1$ $f(x)=k$ has 4 distinct real roots

c)



vertically $N \cos \theta = F \sin \theta + mg$ ①

Horizontally $N \sin \theta + F \cos \theta = \frac{mv^2}{r}$ ②

① x sin θ $N \cos \theta \sin \theta = F \sin^2 \theta + mg \sin \theta$ ③

② x cos θ $N \cos \theta \sin \theta = -F \cos^2 \theta + \frac{mv^2}{r} \cos \theta$ ④

③ - ④ $0 = F + mg \sin \theta - \frac{mv^2}{r}$

$F = \frac{mv^2}{r} \cos \theta - mg \sin \theta$ #

① x cos θ $N \cos \theta = F \sin \theta \cos \theta + mg \cos \theta$ ⑤

② x sin θ $N \sin \theta = -F \cos \theta \sin \theta + \frac{mv^2}{r} \sin \theta$ ⑥

⑤ + ⑥ $N = m \left[g \cos \theta + \frac{v^2}{r} \sin \theta \right]$ #

since $\cos^2 \theta + \sin^2 \theta = 1$

When $F=0$

$\frac{mv^2}{r} \cos \theta = mg \sin \theta$

$\frac{v^2}{1000 \times 10} = \tan \theta$ ($g=10$)

$\frac{v^2}{10000} = \frac{1}{100}$

$r = 1000 \text{ m}$

$v^2 = \frac{10000}{100} = 100$

$v = 10 \text{ m/s}$ #

Question 3

a) $\int_0^4 \frac{dx}{3+\sqrt{x}} =$ $\sqrt{x} = u-3 \quad u > 3$
 $x = (u-3)^2$

$= \int_3^5 \frac{2(u-3) du}{u}$ $dx = 2(u-3) du$

$= \int_3^5 2 du - \int_3^5 \frac{6}{u} du$

$= \left[2u \right]_3^5 - 6 \left[\ln u \right]_3^5$

$= 2 \times 2 - 6 \ln \frac{5}{3}$ #

MATHEMATICS Extension 2: Question...3...

Suggested Solutions	Marks	Marker's Comments
a) If $I = \int_0^4 \frac{dx}{3+\sqrt{x}}$, substitute $u^2 = x$ ($u > 0$) $"2u du = dx"$ When $x=4$, $u=2$ $x=0$, $u=0$ $I = \int_0^2 \frac{2u du}{3+u}$ $= \int_0^2 \frac{2u+6-6}{u+3} du$ $= \int_0^2 \left(2 - \frac{6}{u+3} \right) du$ $= [2u - 6 \ln(u+3)]_0^2$ $= 4 - 6 \ln 5 + 6 \ln 3$ $= \underline{\underline{4 - 6 \ln \left(\frac{5}{3} \right)}}$	1	
b) i) $\alpha + \beta + \gamma = -\frac{b}{a}$ $= -1$ ii) $-(\alpha + \beta) = \gamma - 1$ (from (i)) Roots are to be: $\alpha - 1, \beta - 1, \gamma - 1$ substitute: $y = x - 1 \Rightarrow x = y + 1$ $(y+1)^2 - (y+1) + 2(y+1) - 1 = 0$ $y^2 + 3y^2 + 3y + 1 - y^2 - 2y - 1 + 2y + 2 - 1 = 0$ $\underline{\underline{y^3 + 2y^2 + 3y - 1 = 0}}$	1	Too many used the long method and got lost in the algebra.

2010 TRIAL

MATHEMATICS Extension 2: Question 3... (cont)

Suggested Solutions	Marks	Marker's Comments
c) i) <u>$N = (4 \cos \theta, 0)$</u> <u>$Q = (4 \cos \theta, 4 \sin \theta)$</u> ii) $\frac{dy}{dx} = 3 \cos \theta, \frac{da}{d\theta} = -4 \sin \theta$ $\therefore \frac{dy}{dx} = \frac{dy/d\theta}{da/d\theta} = \frac{-3 \cos \theta}{-4 \sin \theta}$ $y - 3 \sin \theta = \frac{-3 \cos \theta}{-4 \sin \theta} (x - 4 \cos \theta)$ $4y \sin \theta - 12 \sin^2 \theta = -3x \cos \theta + 12 \cos^2 \theta$ <u>$3x \cos \theta + 4y \sin \theta = 12$ (or $x \cos \theta + y \sin \theta = 4$)</u> iii) <u>$4x \cos \theta + 4y \sin \theta = 16$ (or $x \cos \theta + y \sin \theta = 4$)</u> <u>$x \cos \theta + y \sin \theta = 4$ (B)</u> iv) Solve: (A) - 3(B) $y \sin \theta = 0$ $y = 0$ ($0 < \theta < \pi/2$ as P is in Ist Q^{ant}) (so $\sin \theta \neq 0$) <u>Comes at $(4 \sec \theta, 0)$ on x axis.</u> v) $ON \cdot OR = 4 \cos \theta \cdot 4 \sec \theta = 16$ This is independent of θ and hence of P and Q.	1 For marks, needed some defining points on E and C. Also N, Q and P(4 cos θ, 3 sin θ) needed to be marked. (You should know where Q is but it was not required. Nor were the tangents for the first mark.) 1 1 1 1 Too many people derived tangent to the circle. Question said WRITE DOWN (for 1 mark only). 1	

MATHEMATICS Extension 2: Question 4		Marks	Marker's Comments
Suggested Solutions a) $I = \int \frac{dx}{\sqrt{x^2 - 4x + 5}} = \int \frac{dx}{\sqrt{1 + (x-2)^2}} \quad (x > 3)$ $= \ln (x-2) + \sqrt{(x-2)^2 + 1} + C$ $= \ln [(x-2) + \sqrt{x^2 - 4x + 5}] + C$		1	2 $\frac{1}{2}$ For $(x-2)$ $\frac{1}{2}$ For $\ln$ and $\sqrt{\dots}$ $\frac{1}{2}$ For C
b) (i) $\frac{1}{\cos x} = \frac{A \cos x}{1 - \sin x} + \frac{B \cos x}{1 + \sin x}$ because $1 \equiv A(1 + \sin x) + B(1 - \sin x)$ Either use x-values or C-coefficients $x = 0, \pi \Rightarrow A + B = 1$ $x = \frac{\pi}{2} \Rightarrow A - B = 0$ $\therefore A$ and $B = \frac{1}{2}$			1 For correct method $\frac{1}{2}$ For each A, B 2
(ii) $\int_0^{\pi/6} \sec x \, dx = \int_0^{\pi/6} \frac{\frac{1}{2} \cos x}{1 - \sin x} + \frac{\frac{1}{2} \cos x}{1 + \sin x} \, dx$ $= \frac{1}{2} \left[-\ln 1 - \sin x + \ln 1 + \sin x \right]_0^{\pi/6}$ $= \frac{1}{2} \left[-\ln \left(\frac{1}{2} \right) + \ln \left(\frac{3}{2} \right) - (-0 + 0) \right]$ $= \frac{1}{2} \ln 3 \quad \checkmark$		1	$\frac{1}{2}$ For $-\ln 1 - \sin x $ $\frac{1}{2}$ For $\sin \frac{\pi}{6} = \frac{1}{2}$ $\ln \left(\frac{3}{2} \div \frac{1}{2} \right)$ 2
c) (i) A is C 1. $\angle CAT = \angle CET = 90^\circ$ 2. CT is common 3. $CA = CE$ equal radii $\therefore \triangle CAT \cong \triangle CET$ (RHS) Note: SSS and SAS requires a derived result (ii) $\angle ATC = \angle ETC = \theta$ (Corresponding angles in congruent triangles equal) $\therefore \angle ATE = 2\theta$ $\therefore \angle TDF = 2\theta$ (Alternate angles equal as $AT \parallel DF$) $\angle DEF = 90^\circ$ (Vertically opposite angles equal from (i)) $\therefore \cos 2\theta = \frac{DE}{CT} = \frac{CE^2 - ET^2}{CT^2}$ $\therefore DE = DF = \sqrt{ET^2 - CE^2}$ qed.		1	1 For correct method $\frac{1}{2}$ For each A, B 2 $\frac{1}{2}$ For $\cos \theta = \frac{ET}{CT}$ $\frac{1}{2}$ For $\cos 2\theta = \frac{CE^2 - ET^2}{CT^2}$ $\left(\frac{1}{2} \right) \sin \theta = \frac{CE}{CT}$

MATHEMATICS Extension 2: Question 4		Marks	Marker's Comments
Suggested Solutions	(iii) Let $\angle ABT = \beta$ In $\triangle ABT$: $\frac{AT}{\sin \beta} = \frac{AB}{\sin \theta}$ i.e. $\frac{AT}{AB} = \frac{\sin \beta}{\sin \theta}$ $\angle TBD = \pi - \beta$ (angle sum at B) In $\triangle TBD$: $\frac{TD}{\sin(\pi - \beta)} = \frac{BD}{\sin \theta}$ $\frac{TD}{\sin \theta} = \frac{BD}{\sin \theta} \Rightarrow TD = BD$ $\therefore \frac{AT}{\sin \theta} = \frac{AB}{\sin \theta} = \frac{TD}{\sin \theta}$ i.e. $TA = AB = BD$ if θ [this is the angle bisector theorem!]	1 $\frac{1}{2}$ $\frac{1}{2}$	of straight angle is π) 3

MATHEMATICS Extension 2: Question...	5
Suggested Solutions	

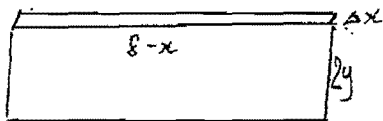
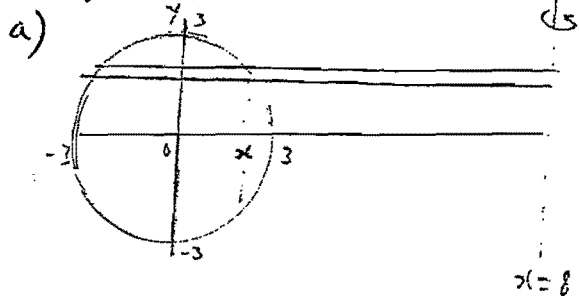
[illegible]

TRIAL 2010 MATHEMATICS Extension 2: Question 6...		
Suggested Solutions	Marks	Marker's Comments
a) $I = \int \frac{dx}{e^x + e^{-x}} = \int \frac{e^x dx}{e^{2x} + 1} \quad (e^x > 0)$ Let $u = e^x$, $\therefore \frac{du}{dx} = e^x$, $\therefore du = e^x dx$ $\therefore I = \int \frac{du}{u^2 + 1} = \tan^{-1} u + k$ $= \tan^{-1}(e^x) + k$	1	
b) i) $I_n = \int_1^e (l \ln y)^n dy$ $I_n = \left[y (l \ln y)^n \right]_1^e - \int_1^e y n (l \ln y)^{n-1} dy$ $I_n = \{e (l \ln e)^n - 1 (l \ln 1)^n\} - \int_1^e n (l \ln y)^{n-1} dy$ $I_n = e - 0 - n \int_1^e (l \ln y)^{n-1} dy$ $I_n = e - n I_{n-1}$	1	Too many short cuts taken with substitution The result is GIVEN and thus must be fully explained
ii) $I_0 = \int_1^e 1 dy = [y]_1^e = e - 1$ $I_2 = e - 2 I_1$ (from above) $= e - 2(e - 2I_0) = e - 2(e - (e - 1))$ $= e - 2$	1	
iii) This is SHM about $x = b$. Low tide 2.0 } Centre of Motion $x = 2.4$ High tide 2.8 } $\therefore b = 2.4$ Period = 13 hrs = $\frac{2\pi}{n}$ $\therefore n = \frac{2\pi}{13}$ Motion at low tide when $t = 0$ $\therefore a = -4$ $\therefore a = -4, b = 2.4, n = \frac{2\pi}{13}$	1/2 1/2	

MATHEMATICS Extension 2: Question 6 (cont)		
Suggested Solutions	Marks	Marker's Comments
c) ii) When $t = 4$, $x = 2.4 - 4 \cos\left(\frac{8\pi}{13}\right) = 2.54 \quad (3 \text{ SF})$ $\therefore$ Depth is 2.54m at 9am	1	Too many people stopped at 2.54 Need units and expressed answer.
iii) The day goes from -5.5 to 5.7 $x = 2.4 - 4 \cos \frac{2\pi t}{13}$ Substitute $x = 2.3$ when $t = T$ $2.3 = 2.4 - 4 \cos \frac{2\pi T}{13}$ $\therefore \cos \frac{2\pi T}{13} = \frac{1}{4}$ $\frac{2\pi T}{13} = 2n\pi \pm \cos^{-1}\left(\frac{1}{4}\right)$ $T = 13n \pm 2.7272$ Only $n = 0$ gives values in allowed range $T = \pm 2.7272 \text{ (hrs)}$ $= \pm 2 \text{ hrs } 44 \text{ min}$ $\therefore$ Times are <u>2.16 am</u> & <u>7.44 am</u>	1	There is nothing in the question to assume tomorrow.
iv) $\frac{dx}{dt} = + \frac{8\pi}{13} \sin \frac{2\pi t}{13}$ $\sin\left(\frac{2\pi t}{13}\right)$ has max value of 1 $\therefore$ Max rate of increase is $\frac{8\pi}{13} \text{ m/hr}$	1	

Question seven

Q7.



Area of cross-section of shell

$$= 2\pi r h = 2\pi (8-x) \Delta x$$

$$\Delta V = 2\pi (8-x) y \Delta x$$

$$\text{Vol} = 4\pi \int_{-3}^3 (8-x) y dx \quad \frac{1}{2}$$

$$= 4\pi \int_{-3}^3 (8-x) \sqrt{9-x^2} dx \quad \frac{1}{2}$$

$$= \pi 32 \int_{-3}^3 \sqrt{9-x^2} dx - 2\pi \int_{-3}^3 x \sqrt{9-x^2} dx$$

Area of

$$= \frac{32}{2} \pi \cdot 3^2 + 2\pi \left[(9-x^2)^{\frac{3}{2}} \right]_{-3}^3$$

$$= 144\pi + 0$$

$$= 144\pi \quad \# 1$$

For B: $xy_1 + yx_1 = 2c^2$ at $x=0$

$$\therefore y = \frac{2c^2}{x_1}$$

$$\text{but } x_1 y_1 = c^2$$

$$\therefore y_1 = 2y_1$$

$$\therefore B = (0, 2y_1) \quad 1$$

Similarly for D: $D = (2x_1, 0) \quad 1$

METHODS TO PROVE ABCD IS A SQUARE worth 2 marks with a conclusion.

Question seven

$$b) \quad c^2 = \frac{a^2}{2} \quad 1$$

$$ii) \quad xy = c^2$$

$$x y' + y \cdot 1 = 0$$

$$y' = -\frac{y}{x}$$

$$\text{At } (x_1, y_1) \quad y' = -\frac{y_1}{x_1}$$

Eq of tangent at (x_1, y_1) is

$$y - y_1 = -\frac{y_1}{x_1} (x - x_1)$$

$$x_1 y - x_1 y_1 = -x y_1 + x_1 y_1$$

$$x_1 y + x y_1 = 2x_1 y_1$$

Now (x_1, y_1) lies on $xy = c^2$

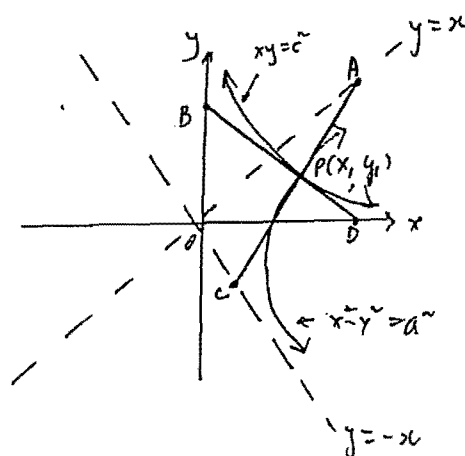
$$\therefore x_1 y_1 = c^2$$

$$\text{Eq tangent: } x_1 y + x y_1 = 2c^2 \quad \# \quad 1$$

$$iii) \quad x^2 - y^2 = a^2$$

$$\frac{x x_1}{a^2} - \frac{y y_1}{a^2} = 1 \quad \# \quad \text{or } x x_1 - y y_1 = a^2 \quad \#$$

iv)



FOR A $x x_1 - y y_1 = a^2$ meet at $y=x$

$$x(x_1 - y_1) = a^2$$

$$\text{But } x_1^2 - y_1^2 = a^2 \therefore (x_1 + y_1)(x_1 - y_1) = a^2$$

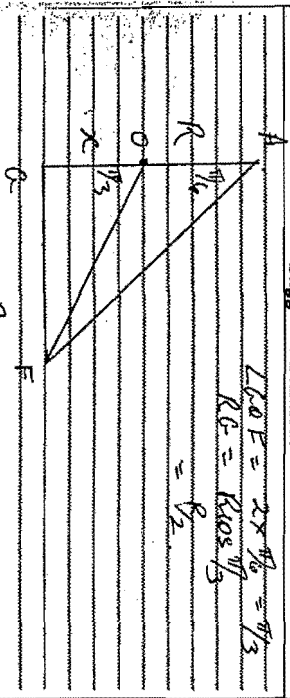
$$\therefore x = \frac{(x_1 + y_1)(x_1 - y_1)}{x_1 - y_1} = x_1 + y_1$$

$$\therefore A = (x_1 + y_1, x_1 + y_1) \quad (x_1 \neq y_1)$$

Similarly for C: $x x_1 - y y_1 = a^2$ meet at $y=-x$

$$\therefore C = (x_1 - y_1, y_1 - x_1) \quad 1$$

MATHEMATICS Extension 2: Question 8...
Suggested Solutions



$$\begin{aligned} \angle COF &= 2 \times \frac{\pi}{6} = \frac{\pi}{3} \\ RC &= R \cos \frac{\pi}{3} \\ &= \frac{R}{2} \end{aligned}$$

(4)

$$\text{Volume} = \int_0^R 4 \tan^2 \theta (R+x)^2 - \pi (R^2 - x^2) dx$$

$$\tan \theta = \frac{1}{\sqrt{3}} \text{ when } \theta = \frac{\pi}{6}$$

$$V = \int_0^{\frac{\pi}{6}} \left[\frac{4}{3} (R+x)^2 - \pi (R^2 - x^2) \right] dx$$

$$= \left[\frac{4}{3} (R+x)^3 - \pi \left[R^2 x - \frac{x^3}{3} \right] \right]_0^{\frac{\pi}{6}}$$

$$= \frac{4}{3} \left[(R^3) - \frac{4}{3} (3R^3) \right] - \pi \left[\left(R^2 - \frac{R^3}{3} \right) - \left(R^2 - \frac{R^3}{3} \right) \right]$$

$$= \frac{4}{3} (8R^3 - 27R^3) - \pi \left(\frac{2R^3}{3} - R^3 + \frac{R^3}{3} \right)$$

$$= \frac{37}{18} R^3 - \frac{5}{24} \pi R^3$$

Marker's Comments

(1) + (1) for
limit values

(1) integration
(1) sub R_2

simplified
(1) answer